

Common Patterns in Time Series

➤ Common Time Series Patterns

1. Trend

- Long-term **upward** or **downward** movement.
- Example: Moore's Law (upward trend in transistor count).

2. Seasonality

- Repeats at **predictable intervals** (daily, weekly, yearly).
- Example:
 - Website usage drops on weekends.
 - Retail spikes during holidays.

3. Trend + Seasonality

- Overall upward/downward movement **with repeating cycles**.
- Example: E-commerce site growing yearly but with seasonal sales peaks.

4. White Noise

- Completely **random** values.
- No useful pattern for prediction.

5. Autocorrelation

- Current value depends on **previous values (lags)**.
- Example: Value = 99% of previous step + occasional spikes.
- **Lag**: Delay in correlation (e.g., lag 1 = immediate step before, lag 50 = 50 steps back).
- **Innovations**: Sudden unpredictable spikes.

6. Mixed Patterns

- Real-world series often have **trend, seasonality, autocorrelation, and noise** combined.

➤ Stationary vs. Non-Stationary

- **Stationary** → Behavior consistent over time → More data = better predictions.
- **Non-Stationary** → Behavior changes (e.g., market crash, tech breakthrough) → Best to train on a **relevant time window**.

➤ Key Insight:

ML models can capture **patterns** (trend, seasonality, autocorrelation), but **noise** is unpredictable. When behavior changes drastically, older data can hurt predictions.
